PART A: 15 OBJECTIVE QUESTIONS [Total mark 15 points]

Answer all the questions. Read each statement carefully. Please answer in Attachment A (page 10).

1. One of the parallel adder types that is based on how it handles a carry value is a _____ carry adder.

   A) generation
   B) look back
   C) override
   D) ripple

2. A decoder uses _____ gate if we want the output to be **HIGH**.

   A) AND
   B) OR
   C) NAND
   D) XNOR

3. Choose the **FALSE** statement about **DEMUX**.

   A) It is also known as a data selector.
   B) A DEMUX is basically the reverse of the multiplexing function.
   C) It takes digital information from one line and distributes it to a given number of output lines.
   D) A decoder can also be used as a DEMUX.

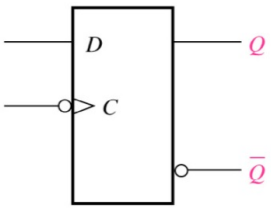
4. 

Figure 1

   Select the **CORRECT** characteristic of the Figure 1 above.

   A) It is a negative edge-triggered flip-flop.
   B) It is a negative edge-triggered latch.
   C) It is a positive edge-triggered flip-flop.
   D) It is a positive edge-triggered latch.

PART B: 4 SUBJECTIVE QUESTIONS [Total mark 85 points]

Question 1 [15 Marks]

Answer the following questions about some functions of combinational logic.

(a) A logic symbol of a parallel adder for summing TWO binary numbers is illustrated in Figure 3. If A = 0011_2 and B = 0110_2 with the carry in, C_0 = 0, complete Table 1 by reproducing it in your answer booklet. Assume pin label 1 represents LSB. [4M]

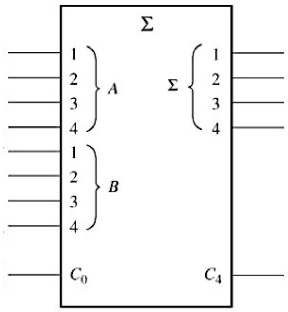


Figure 3

Table 1

Pin Label (i)	Input		Carry Out	Sum
	A_i	B_i	C_i	Σ_i

(b) Draw the logic circuit for a binary decoder to detect the binary code 10010 which produces an active-LOW output. Assume the inputs are represented as $A_4A_3A_2A_1A_0$. [3M]

(c) Given an active HIGH 7-segment display in Figure 4 displaying a digital number from a decoder that based on the BCD value, fill in Table 2. Reproduce Table 2 in your answer booklet. [4M]

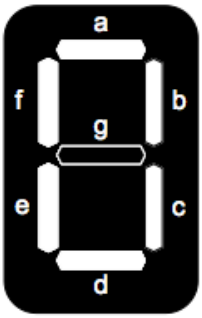


Figure 4

Table 2

BCD Input				Segment Output							Display
A	*B*	*C*	*D*	*a*	*b*	*c*	*d*	*e*	*f*	*g*	
											4
											7
1	0	0	1								9

(d) Figure 5 (in page 11) shows the waveforms of data-input D_i and data-select S_i applied to a multiplexer. Complete Figure 5 by drawing the waveform output of ***X*** in relation to the inputs. Label each segment of output ***X*** with the correct data input selection. [4M]

Question 2 [15 Marks]

(a) Figure 6 shows a basic circuit of a latch. Answer the following based on the circuit.

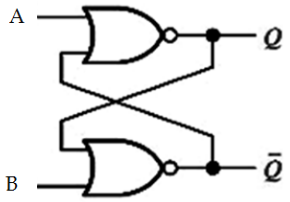


Figure 6

(i) Complete the following truth table with initial values of Q and $\overline{Q}$ are bit 1 and 0. [4M]

Input		Output		State
A	B	Q	$\overline{Q}$	
0	0			
0	1			
1	0			
1	1			

(ii) What type of latch is represented by the circuit in Figure 6? [1M]